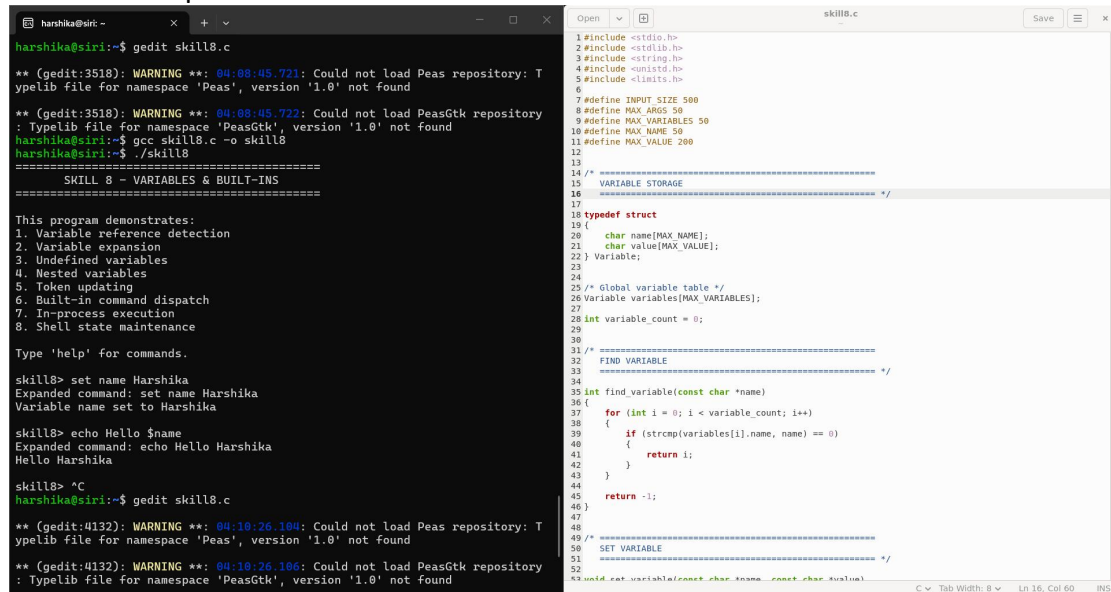


Skill - 8

1. Variable expansion



The screenshot shows a terminal window on the left and a code editor on the right. The terminal window displays the execution of the 'skill8' program, which implements a simple shell. The code editor shows the source code of 'skill8.c'.

```
harshika@siri:~$ gedit skill8.c

** (gedit:3518): WARNING **: 04:08:45.721: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:3518): WARNING **: 04:08:45.722: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gcc skill8.c -o skill8
harshika@siri:~$ ./skill8

=====
SKILL 8 - VARIABLES & BUILT-INS
=====

This program demonstrates:
1. Variable reference detection
2. Variable expansion
3. Undefined variables
4. Nested variables
5. Token updating
6. Built-in command dispatch
7. In-process execution
8. Shell state maintenance

Type 'help' for commands.

skill8> set name Harshika
Expanded command: set name Harshika
Variable name set to Harshika

skill8> echo Hello $name
Expanded command: echo Hello Harshika
Hello Harshika

skill8> ^C
harshika@siri:~$ gedit skill8.c

** (gedit:4132): WARNING **: 04:10:26.104: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:4132): WARNING **: 04:10:26.106: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4 #include <unistd.h>
5 #include <limits.h>
6
7 #define INPUT_SIZE 300
8 #define MAX_ARGS 50
9 #define MAX_VARIABLES 50
10 #define MAX_NAME 50
11 #define MAX_VALUE 200
12
13
14 /*
15  VARIABLE STORAGE
16  ===== */
17
18 typedef struct
19 {
20     char name[MAX_NAME];
21     char value[MAX_VALUE];
22 } Variable;
23
24
25 /* Global variable table */
26 Variable variables[MAX_VARIABLES];
27
28 int variable_count = 0;
29
30
31 /*
32  FIND VARIABLE
33  ===== */
34
35 int find_variable(const char *name)
36 {
37     for (int i = 0; i < variable_count; i++)
38     {
39         if (strcmp(variables[i].name, name) == 0)
40         {
41             return i;
42         }
43     }
44
45     return -1;
46 }
47
48
49 /*
50  SET VARIABLE
51  ===== */
52
53 void set_variable(const char *name, const char *value)
```